INCH-POUND
A-A-59397A (AF)
15, November 2024
SUPERSEEDING
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COMMERICAL ITEM DESCRIPTION

WELDING MACHINES, ARC, DC, GENERATOR, CONSTANT CURRENT, INTERNAL COMBUSTION ENGINE DRIVEN

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes the government acquisition requirements for DC generator arc welding machines driven by internal combustion engines to be used for portable arc welding applications for use in Air Force (AF) Metals Technology (MT) shop installations. AF MT shops serve as the organic and local manufacturing / fabrication resource across the enterprise (150+ locations).

1.1 Intended use. The engine-driven arc welding machines covered by this CID are intended for manual welding used in operations that require the efficient and permanent joining of similar, and in some cases, dissimilar metal pieces. The machines are intended for use in Air Force Metals Technology shops via Air Force Specialty Code (AFSC) 2A7X1.

2. CLASSIFICATION. The arc welding machines shall be of the following types per Table 1.

TABLE I. Arc welding machine classifications.

Drive Power	Type	Auxiliary Power Peak/Continuous (watts)	Min-Max amps (A)	Cooling
Gasoline Engine	01	8000 /6500	30-210	Air
	02	11000/9500	20-265	Air
	03	12000/10500	20-330	Air
	04	12000/10500	20-325	Air
Diesel Engine	05	8000 /6500	30-210	Liquid
	06	12000/10500	20-330	Liquid

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3. SALIENT CHARACTERISTICS

3.1 General requirements. The portable arc welding machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. The machine shall be designed and manufactured in accordance with the American National Standards Institute/National Electrical Manufacturers Association (ANSI/NEMA) EW-1.

3.2 Components. The arc welding machine shall consist of an internal combustion engine, a DC electrical power generator capable of delivering constant direct current at levels suitable for arc welding, a frame, an enclosure, output terminals, and operating controls. Additionally, the welding machine shall include, at a minimum the following components.

3.3 Enclosure. The portable arc welding machine enclosure shall be manufactured to withstand the rough handling normally encountered in construction sites and industrial/welding shop areas. The enclosure shall be constructed of metal materials (sheet metal, cast, wrought), and the design shall be resistant to impact, combustibility, temperature distortion, corrosion, and thermal aging experienced during normal service use and within non-environmentally/climatically controlled machine-shop locations at various Air Force locations. Ventilation holes shall be covered with screens or shall be small enough to prevent the entry of debris. All electrical connection points and machine controls shall be clearly labeled.

3.4 Machine Frame. The machine frame shall include a lifting eye at the normal center of balance at the top of the machine. The lifting eye shall be capable of supporting two and one-half times the weight of the machine and shall be mounted so that either base-mounted or skid-mounted machines will retain alignment during lifting operations.

3.5 Performance. The output rating and performance capabilities of each machine shall comply with the Class I requirements of National Electrical Manufacturers Association (NEMA) Standard EW-1, "Electric Arc Welding Power Sources". Each machine shall be rated for a 100 percent duty cycle and shall meet the performance requirements of ISO 1708 when tested in accordance with sections 5.3 through 5.3.3.

3.5.1 Open Circuit Requirements. The welding machine open circuit voltage shall meet the requirements of NEMA EW-1.

3.5.2 Operating Temperature Limits. The operating temperature limits of the welding machine generator and of all machine parts subject to damage by excessive temperature shall meet the requirements of NEMA EW-1.

3.6 Engine. The arc welding machine prime power source shall be a standard internal combustion engine with the following specifications per Table II, at a minimum.

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TABLE II. Engine Specifications.

Type	Horsepower (HP) per 3600 RPM	Engine Type	Fuel Capacity	Oil Capacity	Standard Shutdown
I	23 hp	Twin-cylinder, 4-cycle overhead valve, industrial air-cooled, gasoline electronic fuel injection (EFI) with electronic throttle body (ETB)	7 gal	2.0 qt	Low oil pressure
II	23.5 hp		11 gal	1.8 qt	
III	27 hp		11 gal	1.75 qt	
IV	23.5 hp		12 gal	1.5 qt	
V	19 hp	3-cylinder, industrial liquid-cooled diesel with 60A alternator and electronic governor. Standard engine block heater.	7 gal	3.4 qt	Low oil pressure, High coolant temperature, Low fuel level
VI	24.8 hp	3-cylinder, industrial liquid-cooled diesel	11 gal	3.9 qt	High coolant temperature, Low oil pressure, Low fuel level

3.7 Output power source. The welding machine electrical power source shall meet the requirements of NEMA EW-1 and shall produce constant-current, self-regulating output power with a drooping volt-ampere characteristic.

3.7.1 Welding Current. The welding current output shall be manually regulated in two steps. A coarse control shall select convenient output current increments through the entire range of the machine. A fine control shall then provide stepless adjustments within the range of each selected increment.

3.7.2 Auxiliary Power Requirements. The machine shall contain an auxiliary power system meeting the requirements of NEMA EW-1. The auxiliary system shall include a 120-volt AC duplex receptacle circuit for lights and power tools and a 230-volt AC receptacle circuit. Both auxiliary circuits shall include automatic overload protection devices.

3.8 Controls. The minimum welding machine controls for Type I thru Type VI are described in Table.111 thru Table V1.

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TABLE III. Controls and display for Type I and Type V portable welding machines.

Control	Display/Function
Air compressor switch	Switch for 20 CFM / 30 CFM setting
Battery charge/weld switch	12 / 24 V battery charge/crank assist
Process control switch	Weld process/battery charge switch
Setup light	Service menu light
Battery charge/crank assist receptacle	12 / 24 V receptacle
Remote 10-pin receptacle	MIG spool gun connection
Menu display/arc control indicator lights	Illuminates menu display / arc control indicator
Maintenance indicator lights	Engine oil/compressor oil change indicator
Engine control switch	Start engine/speed selection/engine stop
Fuel level indicator lights	Illuminates fuel level
Glow plug light (Type V only)	Illuminates when glow plugs are energized
120/240 V receptacles w/ circuit breakers	120/240 V plug-in

TABLE IV. Controls and display for Type II portable welding machines.

Control	Display/Function
Digital display	Weld process/parameters/fuel level
Battery charge/crank assist receptacle	12 / 24 V receptacle
Pushbutton main control knob	Adjust amperage/voltage
Standard 10-pin receptacle	MIG spool gun connection
Pushbuttons	Weld process selection/menu
Engine control switch	Start engine/speed selection/engine stop
Auto-set button	Pre-defined process parameters settings
120/240 V receptacle w/ circuit breaker	120/240 V plug-in

TABLE V. Controls and display for Type III portable welding machines.

Control	Display/Function
Push buttons with digital display	Weld process/parameters, fuel level, maintenance & machine use information
Standard 14-pin receptacle	Connection of wire feeder
Battery charge/crank assist receptacle	12 / 24 V receptacle/power voltage switch
Main control knob pushbutton	Amperage/Voltage adjust
Standard 10-pin receptacle	MIG spool gun connection
Engine control switch	Start engine switch
Auto-Set button	Process pre-defined parameters
120/240 V receptacles w/ circuit breakers	120/240 V plug-in

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TABLE VI. Controls and display for Type IV and Type VI portable welding machines.

Display	Features
Digital Display	Weld process/parameters, fuel level, maintenance/machine use information
Standard 14-pin receptacle	Wire feeder/remote control connection
Battery charge/crank assist	12/24 V receptacle and power/voltage switch
Pushbutton main control know	Amperage/voltage adjustment
Standard 10-pin receptacle	MIG spool gun connection
Standard push buttons	Weld process, menu, machine feature selection
Engine control switch	Engine idle/run adjustment
Auto-Set button	Process specific/pre-defined parameters
120/240 V receptacle w/circuit breakers	120/240 V plug-in
240 V receptacle w/circuit breaker	240 V plug-in

3.9 Cables. A shielded metal arc welding (SMAW) electrode cable (50 ft) with holder, and a work cable (50 ft) with heavy-duty ground spring-style clamps shall be provided.

3.10 Portability. The engine-driven arc welding machine shall be mounted on an over the road trailer. The trailer shall have a minimum load capacity of 1400 lbs., manufactured of heavy-duty steel tubular frame, heavy duty axle with roller bearing hubs and leaf-spring suspension. The trailer shall include a jack stand, fenders, lights, and dual hitch to accommodate a 2 inch ball hitch and 3 inch lunette eye. The trailer shall also have a means of attaching a fire extinguisher and storing documents, i.e. owner's manual, and job site specific paperwork.

3.11 Safety and health requirements. The manufacturer shall ensure that the portable arc welding machine and all equipment and accessories used on the machine shall be in compliance with American Welding Society (ANSI/AWS Z49.1 and Occupational Safety and Health Administration (OSHA) standards, the OSHA standards shall apply/. Covers, guards, or other safety devices shall be provided for all parts of the system that present safety hazards. The manufacturer shall ensure that the machine is intended for manual use. A machine having a power source with an open-circuit voltage greater than those listed in NEMA EW-1, section 5.3.2 shall not be provided as a manual system.

3.12 Systems of units. The U.S. Customary System of Units (US) shall be used to graduate measuring and indicating devices.

3.13 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified, the nameplate shall contain the following information:

a. Manufacturer's name.

b. Machine model designation.

c. Power input (volts, total amperes, phase, and frequency).

d. Rated load volts and amperes.

e. Maximum open circuit voltage.

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f. Duty cycle at rated load.

g. Contract number or order number.

h. National stock number or plant equipment code.

i. Date of manufacture.

4 REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environment Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of material hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operations/use of the item.

5 QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The engine-driven arc welding machines provided shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same system offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The engine-driven arc welding machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified, the welding machine shall be tested in accordance with sections 5.3.1 through 5.3.3.

5.3.1 Test setup and conditions. The manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration relevant to each test.

5.3.2 Operational test. The engine-driven arc welding machine shall be operated in accordance with the manufacturer's standard operating procedures. Proper operation of all controls and features shall be verified during the test period.

5.3.3 Performance tests. The performance tests shall be conducted in accordance with the following sections of NEMA EW-1:

a. Drop test – section 3.8.2 of NEMA EW-1.

b. Output test – section 5 of NEMA EW-1.

c. Temperature rise test – section 6 of NEMA EW-1.

d. High-potential test – section 7 of NEMA EW-1.

e. High-frequency generator test – section 10 of NEMA EW-1.

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5.4 Acceptance. Preliminary and final acceptance tests shall be conducted at the manufacturer's site or at a location agreed upon by the Air Force (AF) Metal Fabrication and Technologies Office (MFTO), Robins AFB, GA. Failure of the machine to meet the tests called out in sections 5.3 through 5.3.3 shall be cause for rejections. AF MFTO are the engineering review and approval authority for all testing resulting from the requirements listed herein.

6.. PACKAGING

6.1 Preservation, packing, and marking. The engine-driven arc welding machine shall be preserved, packed, and marked as specified in the contract or order.

7. NOTES

7.1 Sources of documents.

7.1.1 Government documents. Copies of specifications, standards, and handbooks are available online at https://assist.dla.mil.

7.1.2 American National Standards Institute (ANSI) documents. Copies of ANSI documents may be obtained at https://ansi.org.

7.1.3 National Electrical Manufacturers Association (NEMA). Copies of NEMA documents may be obtained at https://www.nema.org/.

7.1.4 Occupational Safety and Health Association (OSHA). Copies of OSHA documents may be obtained at https://www.osha.gov/.

7.1.5 Federal Acquisition Regulation (FAR) Copies of FARS may be obtained at https://www.acquisition.gov/browse/index/far/.

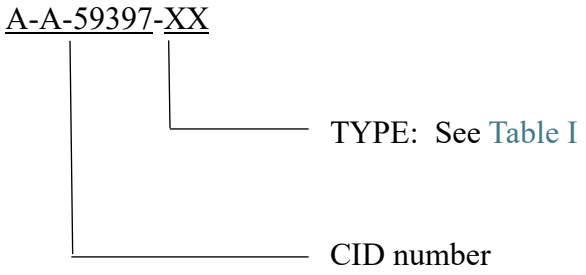
7.1.6 Environmental Protection Agency (EPA). Copies of EPA documents may be obtained at https://www.epa.gov/

7.1.7 National Institute of Standards and Technology (NIST). Copies of NIST documents may be obtained at https://www.nist.gov.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type of engine-driven arc welding system required (Table I).

7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



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7.4 Key Words.

Portable Welder.
Portable Welding Machine.
Mobile Welding Machine.
Mobile Welder.
Welder, Engine Driven.
Welding Machine, Engine Driven.

MILITARY INTERESTS:

Custodian:

Air Force – 184

Reviewer:

DLA – GS

Preparing Activity:

Air Force – 184

Agent:

Air Force – 110

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